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Measurement invariance testing in partial least squares structural equation modeling

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ABSTRACT

When using structural equation modeling, comparison across time or groups can be misleading if measures are not invariant. Partial least squares structural equation modeling (PLS-SEM) is a method widely used in business research, but its ability to test for measurement invariance is limited. This study introduces a comprehensive approach for measurement invariance testing in reflective measurement models in PLS-SEM. The methodology diverges from the traditional measurement invariance of composite models (MICOM) approach and expands the possibilities of measurement invariance testing in three areas: 1) providing statistical tests to validate the comparison of latent means across groups; 2) measurement invariance testing in longitudinal studies; and 3) the ability to simultaneously assess measurement invariance across multiple groups. Additionally, this study proposes a strategy to address measurement invariance rejections in large-sample studies. The paper offers guidelines for the MI tests, and an empirical example illustrates their utility in facilitating experimental approaches in PLS-SEM.

1. Introduction

Measurement invariance (MI), in general, refers to the hypothesis that a construct is measured similarly across groups or occasions (Widaman et al., 2010). If the measures are not invariant, we cannot be sure that differences between groups or occasions are due to differences in the construct or due to differences in measuring the construct. In fact, group comparisons can be meaningless without establishing measurement invariance (Jeong & Lee, 2019), and the group differences may be due to methodological artifacts rather than the substantive nature of the studied phenomenon (Leitgöb et al., 2023). Therefore, it is generally recommended not to compare non-equivalent measures (Solarino & Buckley, 2023; Steenkamp & Baumgartner, 1998; Vandenberg & Lance, 2000). Yet, despite the significance of validating MI, a recent literature review by Cheah et al. (2023) found that out of 378 articles utilizing partial least squares structural equation modeling (PLS-SEM) multigroup analysis, 61 % neglected to test for MI. Hence, most studies fail to verify the legitimacy of any group comparisons.

Comparison between groups is regularly conducted in structural equation modeling (SEM) (Alrawad et al., 2023; Hsieh et al., 2008; Keil et al., 2000; Sharma et al., 2020; Zhu et al., 2006) and so are comparisons across time periods (Qureshi & Compeau, 2009; Venkatesh et al., 2017). Two widely used SEM methods are covariance-based SEM (CB-SEM) and PLS-SEM (Urbach & Ahlemann, 2010). MI tests in CB-SEM are well developed and continue to evolve with new methodological advancements (see, for instance, Solarino & Buckley, 2023; Steenkamp & Maydeu-Olivares, 2021, 2023). The scientific impact of works, such as Steenkamp and Baumgartner (1998) and Vandenberg and Lance (2000), emphasize the widespread application of MI tests within CB-SEM.

Even though researchers predominantly use reflectively measured constructs in PLS-SEM (Hair et al., 2017a), there is currently not an MI methodology explicitly testing the parameters of the reflective measurement model using the well-developed MI principles applied in CB-SEM. For this reason, and in line with calls for MI tests in PLS-SEM (Ringle et al., 2012) and longitudinal studies (Richter et al., 2016), this study develops a general procedure for testing MI in PLS-SEM on reflective constructs. Currently, applied researchers using PLS-SEM almost exclusively utilize the measurement invariance of composite models (MICOM) approach (Henseler et al., 2016) for MI testing. In comparison to MICOM, the methodology proposed in this study offers three new capabilities for MI testing: it can test the validity of comparing latent variable mean differences across groups; it can be utilized in longitudinal studies; and it can simultaneously test for MI across multiple groups. In contrast to MICOM, the method proposed in this study does not offer a test for validating comparisons of path coefficients when constructs are measured formatively.

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The novel MI test presented in this paper enables researchers to compare regression coefficients and latent variable means across groups or time periods.[1] The method is particularly beneficial for business research, which often relies on PLS-SEM (Ahrholdt et al., 2019; Camisón & Villar-López, 2014; Cao et al., 2021; Coltman et al., 2008; Hult et al., 2018; Méndez-Suárez & Monfort, 2020; Schubring et al., 2016). For example, Steelman et al. (2014) examined the technology acceptance model and the expectation–disconfirmation theory, when sampling from online crowdsourcing markets in contrast to more traditional sampling frames. Using CB-SEM MI tests they found that the constructs based on samples from online crowdsourcing markets in the US was measured similarly to constructs obtained from traditional sampling frames.[2] Hence, their results show that it is plausible to compare results in the technology acceptance model or expectation–disconfirmation theory between these samples. However, when the study used PLS-SEM, Steelman et al. (2014) were unable to replicate the MI tests and instead compared the average latent construct scores across the groups. By using the MI tests proposed here, they would have been able to use PLS-SEM while adhering to well-developed MI principles.

As this present study considers PLS-SEM, latent variables are estimated as composites. That is, an unobserved latent variable is approximated by a weighted sum of its indicators. The composite nature of PLS-SEM refers to its approach in approximating latent variables, rather than designating a measurement model as either reflective or formative (Sarstedt et al., 2016). The composites are used in the estimation of loadings, which in turn are utilized for assessing, e.g. internal consistency reliability, indicator reliability, and average variance extracted in the reflective measurement model (Hair et al., 2021). In a similar vein, when assessing MI in the reflective measurement model, the author takes point of departure in the reflective measurement model as estimated by PLS-SEM.

The proposed method aligns with the traditional multigroup analysis in PLS-SEM found in works such as Ahuja and Thatcher (2005), Enns et al. (2003), Hsieh et al. (2008), and Zhu et al. (2006). Additionally, it facilitates the integration of PLS-SEM into longitudinal and experimental contexts, mirroring CB-SEM practices as showcased by, for example, Ployhart and Oswald (2004). This resonates with calls for longitudinal analysis in PLS-SEM (Richter et al., 2016), and provides an opportunity for regular PLS-SEM users to broaden the range of hypotheses they investigate.

The rest of the article is as follows: Section 2 relates the existing MI PLS-SEM literature to the procedure suggested in this paper. Section 3 outlines the reflective measurement model as estimated by PLS-SEM. In Section 4, taking the MI testing steps detailed by sources such as Vandenberg and Morelli (2016), Van Herk and Goldman (2022), and Boer et al. (2018) as a foundation, each of these tests is adapted to the PLS-SEM framework. Section 5 presents a simulation study, examining the power and Type I error of the proposed test across different simulation set-ups. The findings are then synthesized into a recommended procedure in Section 6. Section 7 illustrates the application of the MI procedure on a dataset consisting of a treatment and a control group, each with two repeated measures. Subsequently, a latent ANCOVA model is employed to gauge the impact of information about data breaches on an individual's perceived risk in online shopping. Section 8 discusses the study's limitations and proposes avenues for future research, and Section 9 concludes.

2. Existing measurement invariance tests in PLS-SEM

To the best of the author's knowledge, there exist two developed approaches, supported by simulation studies, for testing MI in PLS-SEM: the MICOM procedure proposed by Henseler et al. (2016); and the invariance test developed in Lamberti and Banet (2017). Before these test procedures, measurement invariance tests in composites were conducted (Hsieh et al., 2008; Ringle et al., 2011), but these advances have been characterized as "… limited or incomplete at best" (Henseler et al., 2016). While the approach by Lamberti and Banet (2017) seems to be rarely used, the MICOM procedure has gained widespread dissemination. MICOM has been used in research on, for example, cultural intelligence (Schlägel & Sarstedt, 2016), tourism (Dedeoglu et al., 2023), employee well-being (Ameen et al., 2023), digitalization (Dahl et al., 2023) and incorporated in methodological guidance (Cheah et al., 2020; Cheah et al., 2023; Matthews, 2017; Sarstedt et al., 2017).

While both MICOM and the approach presented in this paper aim to test measurement invariance in composite models, there are significant differences to note: 1) the MICOM procedure is limited to testing two groups at a time, whereas the method in this study accommodates an unrestricted number of groups; 2) MICOM is not designed to handle repeated observations on the same individuals, whereas this study's methodology, as demonstrated through simulations, can accommodate longitudinal data; 3) the current literature lacks results for MICOM regarding power and Type I error, while the procedure proposed in this paper demonstrates satisfactory performance in these areas; 4) the approach outlined in this paper provides a test for assessing the legitimacy of comparing latent variable means between groups, a feature not available using MICOM. Surprisingly, a part of the MICOM procedure involves testing whether latent means are equal across groups but does so without testing whether such a comparison is meaningful from a MI perspective. Section 4 provides clarification of the necessary level of measurement invariance for comparing latent variable means across groups; 5) in contrast to MICOM's approach of using a correlation test between two auxiliary composites to assess the feasibility of comparing structural coefficients across groups, the method presented here adheres to established MI principles, as detailed in sources like Steenkamp and Baumgartner (1998). That is, the equality of loadings and intercepts are tested across groups, offering interpretability aligned with the conceptual underpinnings of MI. For a detailed interpretation of the measurement model from a MI perspective, refer to Section 4; 6) while MICOM is applicable to both reflective and formative measurement models, the approach discussed here is specifically tailored for reflective measurement models.

In contrast to MICOM and the procedure outlined in this study, Lamberti and Banet (2017) test MI for an entire model across all groups. Thus, the researcher avoids testing MI for each construct in the model. While the availability of a global MI test is beneficial, solely relying on such a test can make it hard for researchers rejecting MI to locate which constructs and indicators are causing the problems. Furthermore, the test statistic in Lamberti and Banet (2017) rests on a normality assumption of the latent variables. This deviates from one of the common motivations for opting for PLS-SEM, specifically its absence of distributional assumptions. Like MICOM, Lamberti and Banet (2017) have designed their test for both reflective and formative measurement models. Unlike the MI test presented here, the test in Lamberti and Banet (2017) was not developed to handle longitudinal data or facilitate the comparison of latent means.

3. The measurement model

In a reflective measurement model, changes in the unobserved latent variable are reflected in the indicators measuring it. The latent variable could, for example, be Innovativeness of a firm (Basco et al., 2020), and

[1] While the comparison of regression coefficients and latent variable means are often the primary objective of MI testing, these tests can hold intrinsic value, as highlighted by a reviewer. For instance, in cultural comparisons, the absence of MI suggests cultural differences in the perception of a construct, which warrant further investigation. Furthermore, while MI is often done within a single study, *meta*-analytic studies can also benefit from MI evaluations (Ringwald et al., 2022).

[2] The traditional sampling frames referred to are students or consumer panel samples.

the indicators be CEO ratings from a questionnaire. Hence, an increase in firm Innovativeness is reflected in the CEO questionnaire ratings (the indicators). The strength of the effect from the latent variable to a given indicator is quantified by the loading. These loadings can differ in size, reflecting the possibility that some indicators are more strongly associated with the latent variable. If, for example, an indicator measuring research & development has a higher loading than an indicator measuring the introduction of new products, then the former indicator is strongest in reflecting Innovativeness.[3] As PLS-SEM is a composite method, it utilizes approximations of unobserved latent variables, by forming a weighted sum of the indicators associated with each of those variables. The weights are found by an iterative algorithm (Tenenhaus et al., 2005). The reflective measurement model, as estimated by PLS-SEM, is estimated as the linear projection of indicator x_k, $k = 1, \ldots, K$, on the latent variable approximation Y:

$$x_k = \alpha_k + \pi_k Y + \varepsilon_k, \tag{1}$$

where π_k, is the loading of indicator k, and ε_k is the error term. α_k denotes the intercept and is not estimated (is set to zero) in PLS-SEM applications using standardized variables. However, in the methodology described in this study, it is important to estimate the intercept when the objective of the MI test is to compare latent variable means across groups.

Tests of MI become important when we consider grouped data. In the rest of the article, the author refers to group membership if there are either different classifications of individuals (e.g. male and female) or the same individuals are observed over multiple time periods. Group membership is denoted by $g = 1, \ldots, G$, where G is the total number of groups. The measurement equation across groups can be written as[4]

$$x_{g,k} = \alpha_{g,k} + \pi_{g,k} Y_g + \varepsilon_{g,k}. \tag{2}$$

If, for example, Innovativeness is measured across three countries (Basco et al., 2020), then $G = 3$, and the latent variable approximations Y_1, Y_2, and Y_3 would be the Innovativeness composites from each of these countries.

Including the latent variable Firm Performance (Basco et al., 2020) in this example, a researcher might test 1) if the effect from Innovativeness to Firm Performance is the same (or differs) across the countries, and 2) if the average score on Innovativeness is the same (or differs) across the countries. However, before such tests can be conducted, the researcher needs to ensure that the latent variables (Innovativeness and Firm Performance) are measured in the same way across the context of the countries. When the latent variables are measured the same way, measurement invariance is achieved. For comparisons of structural coefficients (e.g. the effect from Innovativeness to Firm Performance), we need to test for *metric invariance*, and for comparisons of the levels of latent variables, we also need to test for *scalar invariance*.

The three commonly used levels of MI are *configural invariance*, *metric invariance*, and *scalar invariance* (Boer et al., 2018; Van Herk & Goldman, 2022; Vandenberg & Morelli, 2016). The following section will describe these MI levels in the context of PLS-SEM composites and Equation (2).

4. Types of measurement invariance using latent variable scores

The first type of measurement invariance is *configural invariance* and is a prerequisite for inferring that the constructs have a similar meaning across groups (Steinmetz et al., 2009), and it must be established before subsequent tests are meaningful (Vandenberg & Lance, 2000). Configural invariance is achieved if the pattern of non-zero loadings is the same across groups (Steenkamp & Baumgartner, 1998) and if the loadings have the same sign (Meredith, 1993).

In essence, there is configural invariance if the same set of indicators consistently relate to the same constructs across groups, with identical signs on their loadings. During PLS-SEM multigroup analysis, the researcher should assess the reflective measurement model in each group separately (Sarstedt et al., 2011). MI is relevant in PLS-SEM multigroup analysis, and the reflective measurement model should therefore be assessed in each group before MI tests are conducted. There exist comprehensive guidelines for assessing the reflective measurement model (Hair et al., 2021), but it is worth mentioning that this assessment involves excluding indicators with zero (or very low) loadings. To have configural invariance, the researcher should, therefore, verify that after the reflective measurement model is assessed in each group, then like constructs are related to the same set of indicators and that corresponding loadings have the same sign. In this way, the assessment of the reflective measurement model in each group is also a natural step in configural invariance. Moreover, these groupwise assessments correspond with the definition of configural invariance when the results are compared across groups. Relating to the example in Section 3, the reflective measurement model needs to be evaluated according to the guidelines in Hair et al. (2021), and show that each country has the same indicators (with the same sign on their loadings) related to Innovativeness. The same should be done for Firm Performance.

The second type of invariance is metric invariance. This needs to be established in order to infer that the construct has the same meaning across groups (Steinmetz et al., 2009) and in order to compare path coefficients between groups. However, it does not allow for comparing the latent means (Harmsen et al., 2019). Metric invariance ensures that the indicators are measured on the same metric (Steinmetz et al., 2009) and are tested under the null hypothesis that the loadings on like indicators are invariant across groups (Vandenberg & Lance, 2000). Hence, if, in this study's example, one wishes to test whether the path coefficients between Innovativeness and Firm Performance are the same across the countries, it is a prerequisite to test whether the loadings are the same across the countries for each latent variable. To exemplify, consider one of the indicators related to Innovativeness (call it indicator A). Indicator A has a loading in each of the three countries, and we should test that these indicators only deviate due to sampling error. The same principle applies to the rest of the indicators of Innovativeness as well as the indicators related to Firm Performance.

Fig. 1 illustrates the concept of metric invariance in PLS-SEM by depicting four regression lines between an indicator and its associated latent variable scores for four different groups. When comparing groups 1, 2 and 4 in Fig. 1, we see that the groups have the same slope coefficient, which implies metric invariance. In contrast, Group 3 has a different slope coefficient than the other groups, and metric invariance is not present in that group compared to the other groups. Hence, a one-unit increase in the latent variable score in Group 3 does not result in the same expected change in the indicator as would have been the result of a one-unit increase in the latent variable score in groups 1, 2, and 4. Therefore, we cannot say that the indicator across the groups responds similarly to changes in the latent variable.

The third kind of invariance is scalar invariance, which ensures that like constructs are on the same scale (Acar et al., 2019), and the difference in the latent score means is adequately reflected in the difference in the indicator means. Scalar invariance is only tested if there is metric invariance, and scalar invariance is a prerequisite if the researcher wishes to conduct latent mean comparisons between groups (Steinmetz et al., 2009). This level of invariance is tested under the null hypothesis that the intercepts of like indicators are identical across groups (Vandenberg & Lance, 2000). In the context of PLS-SEM, estimation is often

[3] Assuming that both indicators are measured on the same scale, or that the indicators are all standardized.

[4] If the latent variable approximation, Y_g, in Equation (2) is substituted by the true unobserved latent variable η_g, the reflective measurement model is $x_{g,k} = \tau_{g,k} + \rho_{g,k}\eta_g + \delta_{g,k}$, where $\tau_{g,k}$ is the intercept, $\rho_{g,k}$ the loading, and $\delta_{g,k}$ the error term. In this model, the predictor specification $E(x_k|\eta) = \tau_k + \rho_k\eta$ is assumed to hold (Wold, 1982). The predictor specification implies the common assumptions in CB-SEM, namely that $E(\delta_k) = 0$ and $cov(\delta_k, \eta) = 0$ (Bollen, 1989).

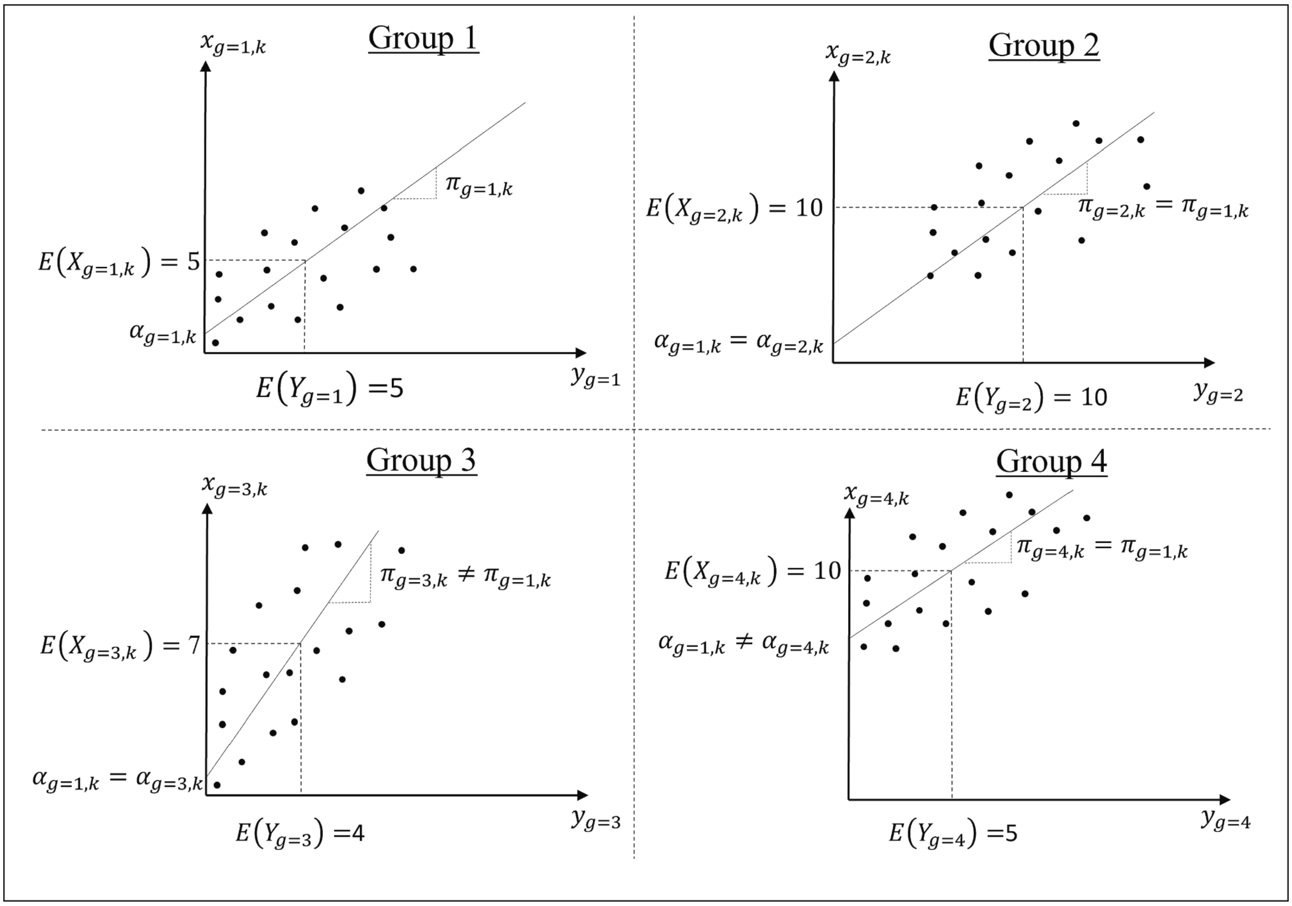

Fig. 1. Metric and scalar (in)variance.
Note: Y_g denotes a composite from group g, and $x_{g,k}$ is indicator k in group g. Furthermore $\alpha_{g,k}$, and $\pi_{g,k}$ are intercept and loading from the PLS-SEM reflective measurement model given as $x_{g,k} = \alpha_{g,k} + \pi_{g,k} Y_g + \varepsilon_{g,k}$.

conducted on fully standardized data. However, when investigating latent means, the means of the indicators (and the latent variables) should not be centered around zero (Hair et al. 2017b; Ringle & Sarstedt, 2016). A prominent example is Fornell et al. (1996), which compares averages of the American Customer Satisfaction Index (ACSI) between sectors and industries.

In the current example, a test for scalar invariance is needed if one wishes to determine if the latent means of, for example, Innovativeness, are the same across the three countries. To compare the latent means, the researcher needs to compute latent variable scores, which are allowed to have a non-zero mean (i.e. the latent variable scores are not demeaned as is typically done in PLS-SEM). Correspondingly, when testing the intercepts of the measurement model (the α's from Equation (2)), one uses non-demeaned latent variables and indicators. Consider again indicator A of Innovativeness; we will have an intercept in the measurement model of indicator A for each of the three countries, as three PLS-SEM are estimated (one for each country). To investigate if indicator A is scalar invariant, one tests if the intercepts of indicator A are equal across the countries. Scalar invariance of Innovativeness implies that each of the indicators of Innovativeness should have the same intercepts across the countries.

By comparing groups 1 and 2 in Fig. 1, we see that both the slope and intercept are the same, which implies scalar invariance. Note that we have scalar invariance while the mean of the latent variable and the indicator are different between the groups. When comparing groups 1 and 3, the intercepts are the same, but the indicator does not exhibit scalar invariance due to the lack of metric invariance. Consequently, when the slopes are not the same between groups, identical intercepts do not ensure that a change in the mean of the latent variable is captured in the same way by the mean of the indicator across groups. Finally, by comparing groups 1 and 4, we see that the mean of the indicator differs, whereas the mean of the latent variable is the same. This difference is reflected by a difference in the intercept between the groups, and there is a lack of scalar invariance.

When testing for MI, one should generally proceed in the order in which the different types of MI are described above. In CB-SEM, the concept of partial measurement invariance (Byrne et al., 1989) is often invoked. Partial measurement invariance refers to the case when only a subset of the indicators is invariant across groups, and the variant indicators are kept in the model when comparing regression coefficient or latent variable means. Here it is recommended to delete invariant indicators from the model as the use of partial MI is still debated (Vandenberg & Morelli, 2016). Section 6 gives a detailed overview of a recommended procedure.

4.1. Testing metric invariance

Following the recommendations of Chen (2008), MI is tested one construct at a time. In the case of different classifications of individuals, separate PLS-SEM models are estimated for each group, and when considering repeated measures, the estimated scores stem from a model where like constructs are connected in the nomological net. By estimating the difference in the loadings between a reference group and the other groups, one can test for metric invariance.

Without loss of generality, the first group is set as the reference group. Let $N = \sum_{g=1}^{G} n_g$ be the total sample size where n_g is the sample size in group g, $g = 1,2,\ldots,G$. When comparing the same individuals over multiple time periods it must hold that $n_g = n$ for all g. The difference between the loading of the first group (the reference group) and any other group is $\pi^d_{g^*,k} = \pi_{g^*,k} - \pi_{g=1,k}$, $g^* = 2,\ldots,G$. Under the null hypothesis of equal loadings, these differences will be zero. According to the configural invariance requirement, there must be the same number of indicators over the different groups. The total number of like indicators for a specific construct is denoted by K. Each indicator is stacked in an $N \times 1$ vector, $x_k = \left(x'_{g=1,k}, x'_{g=2,k}, \ldots, x'_{g=G,k}\right)'$. PLS-SEM computes the latent variables with a standard deviation of one in each group. To enable comparisons across groups, the latent variables are furthermore scaled such that the vector, $Y = \left(Y'_{g=1}, \ldots, Y'_{g=G}\right)'$, has a standard deviation of one. The latent variable scores are arranged in an $N \times G$ matrix according to

$$Y^M = \begin{bmatrix} Y_{g=1} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ Y_{g=2} & Y_{g=2} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} \\ Y_{g=3} & \mathbf{0} & Y_{g=3} & \mathbf{0} & \cdots & \mathbf{0} \\ \vdots & \mathbf{0} & \mathbf{0} & \ddots & \mathbf{0} & \vdots \\ \vdots & \vdots & \vdots & \mathbf{0} & \ddots & \mathbf{0} \\ Y_{g=G} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & Y_{g=G} \end{bmatrix} \tag{3}$$

The first column in Y^M is a vector of all latent variable scores because Group 1 is the reference group. In the case that Group 2 is used as the reference group, the vector of all latent variable scores will be in Column 2, and only $Y_{g=1}$ as non-zero in Column 1. Using Y^M allows the specification of the regression equation utilized to test for metric invariance,

$$\begin{bmatrix} x_{k=1} \\ x_{k=2} \\ \vdots \\ x_{k=K} \end{bmatrix} = \begin{bmatrix} Y^M & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & Y^M & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & Y^M \end{bmatrix} \cdot \begin{bmatrix} \pi^A_{k=1} \\ \pi^A_{k=2} \\ \vdots \\ \pi^A_{k=K} \end{bmatrix} + \boldsymbol{\nu} \tag{4}$$

where $\pi^A_k = \left[\pi_{g=1,k}, \pi^d_{g=2,k}, \ldots, \pi^d_{g=G,k}\right]'$, $k = 1,\ldots,K$, is a vector of coefficients, and $\boldsymbol{\nu}$ is an $NG \times 1$ vector of error terms. $\pi_{g=1,k}$ will be the PLS-SEM loading on indicator k in Group 1, and $\pi^d_{g^*,k}$, $g^* = 2,\ldots,G$, will be the difference in the PLS-SEM loadings between Group 1 and Group $g^* = 2,\ldots,G$. If indicator k is problematic in terms of MI, it is likely to have large coefficients on $\pi^d_{g^*,k}$. If all $\pi^d_{g^*,k}$, $g^* = 2,\ldots,G$, are large and similar in magnitude, it implies that the indicator in Group 1 might be problematic in terms of MI.

The null hypothesis of equality of all loadings across groups is,

$$\pi^D_k = \mathbf{0}, \quad k = 1,\ldots,K \tag{5}$$

where $\pi^D_k = \left[\pi^d_{g=2,k}, \ldots, \pi^d_{g=G,k}\right]'$. As an example, having two indicators and two groups, the null hypothesis would be $\pi^d_{g=2,k=1} = \pi^d_{g=2,k=2} = 0$.

The Wald statistic for testing multiple restrictions is used to test the hypothesis of metric invariance. That is, by stacking in a $(G-1)K \times 1$ vector according to.

$\pi^D = \left[(\pi^D_{k=1})', (\pi^D_{k=2})', \cdots, (\pi^D_{k=K})'\right]'$, we can test $H_0: R\pi^D = r$, where R is a $(G-1)K \times (G-1)K$ identity matrix, and r is a $(G-1)K \times 1$ vector of zeroes. The Wald statistics have the form,

$$W = \left(R\hat{\pi}^D - r\right)'\left(R\hat{V}R'\right)^{-1}\left(R\hat{\pi}^D - r\right), \tag{6}$$

where $\hat{V}$ is an estimate of the covariance matrix of π^D [5]. Instead of relying on an asymptotic distribution, the author follows the standard PLS-SEM practice of significance testing by bootstrapping (Hair et al., 2021). Specifically, the test statistics in equation (6) is bootstrapped for inference purposes.

It is also possible to test individual loading differences using Equation (4). However, in doing so one should restrict the null hypothesis in Equation (5) to only include the indicator tested for invariance. For example, to test the second indicator for metric invariance, one would retain the structure of Equation (4), but the null hypothesis will be $\pi^D_{k=2} = 0$.

The proposed test is substantiated with a range of simulation results, presented in Section 5, which indicate that the procedure behaves well under the null hypothesis. Specifically, for the simulated models, the probability of incorrectly rejecting a true null hypothesis (Type I error) corresponds to a predefined significance level. In the longitudinal case, panel bootstrapping is utilized to account for possible correlations within individuals. It is also shown how the power of the metric invariance test rises to one as a function of the sample size and as the difference in loadings between groups gets larger.

4.2. Testing scalar invariance

The purpose of testing for scalar invariance is to be able to compare latent variable means between groups. The test of scalar invariance should therefore allow the group specific means to differ. The author identify the mean structure of the measurement model by means of an identification strategy that is often used in CB-SEM. More specifically, the rationale of a marker indicator (Brown, 2015; Cheung & Rensvold, 1999) is adopted. When testing for scalar invariance, the marker indicator is an indicator that is assumed a priori to have an invariant intercept between groups. The marker indicator together with the use of a reference group can identify differences of intercepts in the measurement model.

The marker indicator should be an item for which the researcher cannot reject the null hypothesis of metric invariance (Vandenberg & Lance, 2000). Similar to, for example, confirmatory factor analysis, difficulties can arise if the researcher inadvertently selects a non-invariant indicator as the marker indicator. One solution to this problem can be to rerun the analysis with different marker indicators (Brown, 2015). For notational simplicity, the marker indicator will be chosen to be the first indicator, $x_{g,k=1}$.

As in the test for metric invariance, the test for scalar invariance also uses a reference group. The difference between the intercept of group $g^* = 2,\ldots,G$ and the first group (reference group) is $\alpha^d_{g=g^*,k} = \alpha_{g=g^*,k} - \alpha_{g=1,k}$, $k = 1,\ldots,K$. Under the null hypothesis of scalar invariance, these deviances will be zero. In order to construct the test equation, a dummy variable matrix is specified as

$$I^M = \begin{bmatrix} 1_{n_{g=1}} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ 1_{n_{g=2}} & 1_{n_{g=2}} & \mathbf{0} & \mathbf{0} \\ \vdots & \mathbf{0} & \ddots & \mathbf{0} \\ 1_{n_{g=G}} & \mathbf{0} & \mathbf{0} & 1_{n_{g=G}} \end{bmatrix} \tag{7}$$

where 1_{n_g} is an $n_g \times 1$ vector of ones. The first column in equation (7) consists entirely of ones because Group 1 is the reference group. If, for example, Group 2 is the reference group, then the second column would consist entirely of ones, whereas the first column would only have ones in the first $n_{g=1}$ rows.

[5] The covariance of π^D is estimated as in the familiar F-test statistics (Cameron & Trivedi, 2005). That is $\hat{V} = s^2\left(Y^{MM'}Y^{MM}\right)^{-1}$ where Y^{MM} is the first matrix on the right-hand side of the equality sign in Equation (4), and s^2 is the estimate of the error variance in the same equation.

To incorporate the marker variable in the testing equation, vectors are introduced, for each $g = 1,\ldots,G$, as follows:

$$\lambda_{g,k=k^*} = x_{g,k=k^*} - \frac{\pi^{res}_{k=k^*}}{\pi^{res}_{k=1}} x_{g,k=1}, \quad k^* = 2,\ldots,K, \tag{8}$$

where $\lambda_{g,k=k^*}$ is a $n_g \times 1$ vector, and π^{res}_k, $k = 1,\ldots,K$, is a slope coefficient that is restricted to be equal over all groups and is obtained from the linear projection of x_k on the vector $Y = \left(Y'_{g=1},\ldots,Y'_{g=G}\right)'$. This restriction is in accordance with the prerequisite condition that metric invariance holds (Nye & Drasgow, 2011). The test equation is,

$$\begin{bmatrix} \lambda_{k=2} \\ \lambda_{k=3} \\ \vdots \\ \lambda_{k=K} \end{bmatrix} = \begin{bmatrix} I^M & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & I^M & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & I^M \end{bmatrix} \cdot \begin{bmatrix} \alpha^A_{k=2} \\ \alpha^A_{k=3} \\ \vdots \\ \alpha^A_{k=K} \end{bmatrix} + \boldsymbol{\vartheta} \tag{9}$$

where $\lambda_k = \left[\lambda'_{g=1,k}, \lambda'_{g=2,k}, \ldots, \lambda'_{g=G,k}\right]'$ and $\alpha^A_k = \left[\alpha^r_{g=1,k}, \alpha^d_{g=2,k}, \ldots, \alpha^d_{g=G,k}\right]'$, $k = 2,\ldots,K$. The first entry in α^A_k will be related to the intercept in Group 1, and the remaining entries, $\alpha^D_k = \left[\alpha^d_{g=2,k},\ldots,\alpha^d_{g=G,k}\right]'$, will be the related to the difference in the intercept between Group 1 and Group $g = 2,\ldots,G$ for indicator $k = 2,\ldots,K$. Appendix A.1 shows how Equation (9) estimates intercept differences. Notice that $k = 1$ does not appear in the stacked vectors in Equation (9). This is due to the choice of indicator one as the marker indicator. If, for example, indicator two is the marker indicator, then the stacked vectors in Equation (9) would have been $\left[\lambda'_{k=1}, \lambda'_{k=3}, \lambda'_{k=4}, \ldots, \lambda'_{k=K}\right]'$ and $\left[\left(\alpha^A_{k=1}\right)', \left(\alpha^A_{k=3}\right)', \left(\alpha^A_{k=4}\right)', \ldots, \left(\alpha^A_{k=K}\right)'\right]'$. Under the null hypothesis of scalar invariance, we have

$$\alpha^D_k \quad = \quad \mathbf{0}, \quad k = 2,\ldots,K. \tag{10}$$

For instance, with three indicators over two groups, the null hypothesis simplifies to $\alpha_{g=2,k=2} = \alpha_{g=2,k=3} = 0$ under the assumption that the marker indicator has invariant intercept over the groups (i.e. $\alpha_{g=1,k=1} - \alpha_{g=2,k=1} = 0$).

To test for scalar invariance, the first step is to estimate π^{res}_k, $k = 1,\ldots,K$ with ordinary least squares (OLS) and insert these estimates into Equation (8) to obtain estimates of $\lambda_{g,k=k^*}$:

$$\widehat{\lambda}_{g,k=k^*} = x_{g,k=k^*} - \frac{\widehat{\pi}^{res}_{k=k^*}}{\widehat{\pi}^{res}_{k=1}} x_{g,k=1}, \quad k^* = 2,\ldots,K. \tag{11}$$

Subsequently, the model given in Equation (9) is estimated with OLS. Finally, the null hypothesis given in Equation (10) is tested with a bootstrapped Wald statistic, as described in Section 4.1. To test for scalar invariance on a single indicator, k, the null hypothesis in Equation (10) is only specified for indicator k. For example, a test of scalar invariance for only the second indicator will have the null hypothesis $\alpha^D_{k=2} = 0$. The proposed test of scalar invariance is substantiated with simulations in Section 5. These simulations show that the scalar invariance test controls the Type I error and has power levels that increase as a function of sample size and differences in intercept.

5. Simulation study

To ensure that researchers can reliably utilize the proposed procedure, its performance is assessed in several Monte Carlo studies by systematically manipulating an expansive set of experimental conditions. False positives (Type I errors) are an important concern in behavioral research (Goodhue et al., 2017). A test statistic with an appropriate Type I error rate is expected to incorrectly reject a true null hypothesis (i.e. Type I error) with a probability of 5 % given a 0.05 significance level (Cameron & Trivedi, 2005). To determine if the proposed test statistics meet this criterion, a range of simulations under the null hypothesis of metric and scalar invariance is conducted. The power, or the probability of correctly rejecting a false null hypothesis, is also investigated. This informs us about the test's capability of identifying potential issues with MI. Each simulation set-up (see Table 1) is calculated with 6000 Monte Carlo repetitions, for the sample sizes 100, 250, 500, and 1000, and variables are drawn from normal distributions.

As MI testing is focused on the measurement model, the inner model is kept simple, with two constructs and a structural coefficient of 0.6 as in Henseler et al. (2016). The number of indicators on each construct is 4, which is close to the median value of the number of indicators used in PLS-SEM (Hair et al. 2017a), and all indicators are reflectively measured. The general simulation model for the two groups is depicted in Fig. 2.

A simulation study for configural invariance is not pursued because this step involves evaluating the reflective measurement model in each group using well-established PLS-SEM criteria, as outlined in Section 4.

The remainder of this section will outline the simulation set-ups, which have been chosen to mirror, as closely as possible, scenarios found in empirical research.

To investigate whether the Type I error probabilities correspond to a predefined 0.05 significance level, five different types of simulations are conducted (see Scenarios 1–5 in Table 1). In each testing scenario, the testing procedure for both metric and scalar invariance is used, and if the tests are adequate, we should observe Type I error rates close to 0.05.

Scenario 1 is a baseline simulation, where the groups have equal latent variable means, equal intercepts, and equal loadings. The other scenarios are variations over the simulation set-up in Scenario 1.

Given that group sizes often differ in real-world settings, it is crucial to ensure that such disparities do not negatively affect the test. Scenario 2, therefore investigates whether difference in group size affects the

Table 1
Overview of experimental factors in the simulation study.

Factor combinations when evaluating Type I error	
Scenario 1	Outer error correlation of 0, the groups have equal means of the latent variables, and all loadings are 0.8. Thereby, the model is the same in both groups.
Scenario 2	As in Scenario 1, but the sample size in Group 2 is half the size of Group 1. This simulation should detect any sensitivity of the Type I error when the group sizes are unequal.
Scenario 3	As in Scenario 1, but the mean of the latent variables in Group 2 is one higher than in Group 1. A proper Type I error in this scenario indicates that the tests of metric and scalar invariance are not affected by different latent variable means as long as the intercepts remain unchanged.
Scenario 4	As in Scenario 1, but with unequal loadings (0.6, 0.7, 0.8, 0.9) within a group, but the same loading pattern across groups. This scenario investigates whether the Type I error changes in response to unequal loadings.
Scenario 5	As in Scenario 1, but with outer error correlation of 0.2. This scenario corresponds to the case of repeated measures on the same individuals, where the outer error terms are correlated. Both panel bootstrapping and cross-sectional bootstrapping are performed. Hence, the Type I error rate between these two bootstrapping methods can be compared.
Factor combinations when evaluating power	
Scenario 6	As in Scenario 1, but the loadings in Group 1 are 0.7, and the loadings in Group 2 range from 0.75 to 0.9. The purpose of this simulation is to investigate how the probability of rejecting a false null hypothesis of metric invariance increases as the difference in loadings increases.
Scenario 7	As in Scenario 6, but with outer error correlation of 0.2. This allows investigating whether outer error correlation affects the power in testing for metric invariance.
Scenario 8	As in Scenario 1, but the intercepts in Group 1 are zero, whereas the intercepts in Group 2 range from 0.1 to 0.4. The purpose of this simulation is to investigate the probability of rejecting a false null hypothesis of scalar invariance.
Scenario 9	As in Scenario 8, but with an outer error correlation of 0.2. This allows investigation into whether outer error correlation affects the power in testing for scalar invariance.

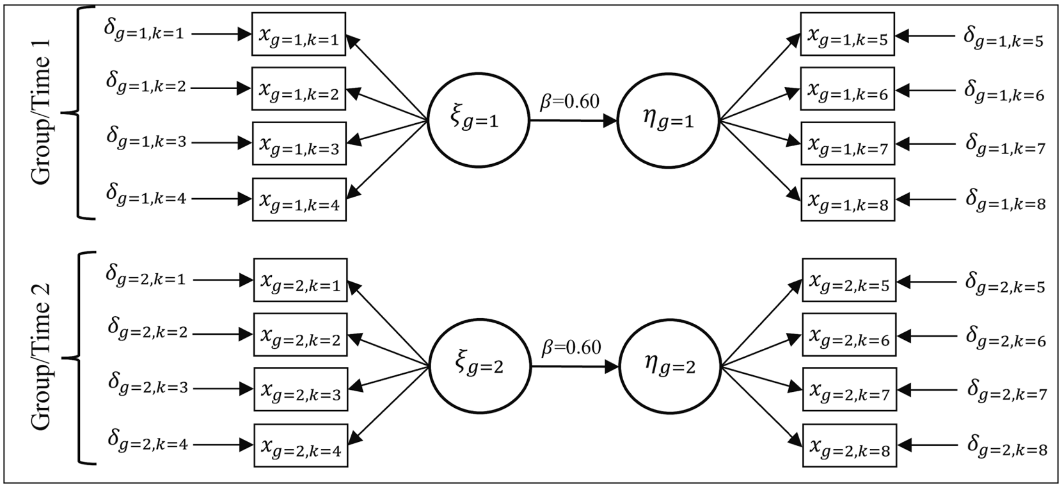

Fig. 2. Population models used in Monte Carlo studies.
ξ and η are latent variables, x denotes indicators, δ is outer errors, and β gives the path coefficients. Subscript g denotes either group (or time when outer errors on like indicators are correlated) and subscript k distinguishes indicators.

proposed tests, by having the sample size in Group 2 being half the size of that of Group 1.

Under scalar invariance, we can compare latent means across groups, even if these means are potentially unequal. Thus, it is important to ensure that varying latent variable means across groups do not influence the test performance. Therefore, Scenario 3 presents simulations where the groups have different latent variable means.

In the metric invariance test, it is assessed whether loadings on like indicators are the same across groups. However, this test still allows for different loadings across indicators within a group. Thus, simulation Scenario 4 varies the loadings within a group, to examine the impact on the proposed tests.

Longitudinal data collected on like indicators often have time-correlated outer errors (Chan, 1998). To investigate the applicability of the proposed methods in a longitudinal setting, Scenario 5 has, therefore, data generated with an outer error correlation of 0.2 between time periods (groups) on like indicators. That is, $cor(\delta_{g=1,k}, \delta_{g=2,k}) = 0.2$ for $k = 1, 2, \ldots, 8$, where $\delta_{g,k}$ is the outer error related to indicator k in time g. Because of this correlation, responses on similar indicators show dependence across time. As bootstrapping assumes independence across individuals, and there are time-correlated errors across responses for an individual, it is therefore appropriate to use panel bootstrapping (Cameron & Trivedi, 2005). This approach involves resampling all time periods, g, for each selected individual within a bootstrap sample. Scenario 5 utilizes both panel and cross-section bootstrapping to examine the impact of the bootstrapping technique on the test's size.

In scenarios 6, 7, 8, and 9 (see Table 1) the simulations focus on the statistical power of the metric and scalar invariance tests. That is, the tests' ability to correctly identify deviations from invariance is investigated. Differences in loadings across groups correspond to a lack of metric invariance. Correspondingly, the difference in loadings is systematically increased across groups in Scenario 6 to investigate how well the metric invariance test detects these differences. It is expected that the proposed test of metric invariance will begin to reject the null hypothesis more frequently as the difference in loadings increases or as the sample size increases. Scenario 7 utilizes the same set-up as Scenario 6, but includes correlated outer errors, and therefore uses panel bootstrapping for inference. Hence, Scenario 7 investigates the power of the test in a longitudinal setting.

In scenarios 8 and 9, the intercepts are different between groups[6] to investigate the power in detecting scalar invariance. We expect to see a rise in power as the intercept difference increases or as the sample gets larger. Scenario 8 is without correlated outer errors, whereas scenario 9 is with correlated outer errors. Again, panel bootstrapping is used in the case of correlated outer errors.

5.1. Results assessment

From Table 2, it is evident that the tests for metric and scalar invariance in general have Type I error rates corresponding to the 0.05 significance level used in the simulation study.

In Scenario 5 (with correlated outer errors), it becomes clear why panel bootstrapping can be beneficial when correlated outer errors are expected. Specifically, when using cross-section bootstrapping, the Type I error rate in testing for scalar invariance is well below the 0.05 level, but using panel bootstrapping, the Type I error rate is close to the pre-specified significance level of 0.05.

Next, the metric and scalar invariance tests are investigated in terms of their ability to reject a false null hypothesis (power). These results are reported in Table 3. In all scenarios, the power level increases with sample size and with deviations from the null hypothesis. Comparing scenarios 6 and 7, the power levels are similar regardless of whether the outer errors are correlated or not.

When testing for scalar invariance, the scenario with correlated outer errors and panel bootstrapping (Scenario 9) has a higher power compared to Scenario 8 with no outer error correlation and cross-section bootstrapping. The higher power levels found in scenarios 8 and 9 compared to scenarios 6 and 7 are not solely due to the fact that the intercepts deviate more between groups than the loadings do. For example, if we compare Scenario 8 with a difference in intercept of 0.2 with Scenario 6 with a difference in loadings of 0.2, we see that the power is almost 0.2 higher for small sample sizes in the test for scalar invariance. In general, there is higher power in detecting scalar invariance than metric invariance.

[6] To simulate under the marker indicator assumption of the scalar invariance test, and similarly to the measurement invariance simulations in Chen (2008), the intercept of the marker indicator is kept equal across groups.

Table 2
Type I error rates based on simulation study.

Scenario	Type of Test	Sample size of largest group			
		100	250	500	1000
1 (Baseline)	Metric invariance	0.049	0.051	0.046	0.049
	Scalar invariance	0.049	0.053	0.053	0.050
2 (Unequal group size)	Metric invariance	0.046	0.048	0.047	0.051
	Scalar invariance	0.047	0.048	0.052	0.052
3 (Different latent means across groups)	Metric invariance	0.047	0.051	0.047	0.055
	Scalar invariance	0.052	0.052	0.051	0.054
4 (Unequal loadings within group)	Metric invariance	0.047	0.050	0.051	0.050
	Scalar invariance	0.046	0.047	0.051	0.046
5 (Outer error correlations)	Metric invariance, cross-section bootstrap	0.046	0.049	0.050	0.050
	Scalar invariance, cross-section bootstrap	0.002	0.002	0.002	0.003
	Metric invariance, panel bootstrap	0.050	0.054	0.053	0.053
	Scalar invariance, panel bootstrap	0.045	0.049	0.048	0.045

Note: The values are the Type I error rates of a given test, using a 0.05 significance level, and are based on 6000 Monte Carlo simulations. The Type I error rates are calculated as $\frac{1}{R}\sum_{r=1}^{R} I[p_r \leq 0.05]$ where $I[\cdot]$ is the indicator function, p_r is the p-value from a test of either metric or scalar invariance, and R is the number of Monte Carlo simulations. Using a significance level of 0.05 implies that an adequate test should give Type I error rates close to 0.05. The numbers in the "Scenario" column corresponds to the simulation designs given in Table 1.

Table 3
Power of tests based on simulation study.

Scenario	Difference in loadings between groups	Sample size in each group			
		100	*250*	*500*	*1000*
6 (Metric invariance)	0.05	0.056	0.081	0.133	0.213
	0.10	0.092	0.212	0.391	0.685
	0.15	0.181	0.440	0.748	0.966
	0.20	0.316	0.694	0.941	0.999
7 (Metric invariance, outer error correlation)	0.05	0.051	0.081	0.125	0.215
	0.10	0.091	0.215	0.407	0.681
	0.15	0.187	0.434	0.745	0.963
	0.20	0.317	0.706	0.946	0.999
	Difference in intercepts between groups				
8 (Scalar invariance)	0.10	0.166	0.341	0.611	0.892
	0.20	0.500	0.889	0.994	1.000
	0.30	0.841	0.996	1.000	1.000
	0.40	0.971	1.000	1.000	1.000
9 (Scalar invariance, outer error correlation)	0.10	0.304	0.654	0.920	0.999
	0.20	0.834	0.998	1.000	1.000
	0.30	0.989	1.000	1.000	1.000
	0.40	1.000	1.000	1.000	1.000

Note: The values are the power of a given test and are based on 6000 Monte Carlo simulations. The power is calculated as $\frac{1}{R}\sum_{r=1}^{R} I[p_r \leq 0.05]$ where $I[\cdot]$ is the indicator function, p_r is the p-value from a test of either metric or scalar invariance in the r'th simulation, and R is the number of Monte Carlo simulations. The numbers in the "Scenario" column correspond to the simulation designs given in Table 1. In scenarios 7 and 9, with outer error correlations, the p-values are based on panel bootstrapping.

5.2. Alternative significance levels

When assessing MI, researchers using large sample sizes are often faced with the problem of rejecting the null hypothesis of measurement invariance for trivial differences in loadings or intercepts (Putnick & Bornstein, 2016). This section presents a promising solution to this problem by introducing the approach of Mudge et al. (2012) in PLS-SEM MI testing. The approach discards the use of a fixed significance level and instead focuses on balancing the Type I and Type II error probabilities in a test. Shifting the focus in this way implies that researchers in general will evaluate the test with lower significance levels in larger samples compared to smaller samples.

In the method proposed in this study, the issue with a fixed significance level materializes when the sample size is increased. By using a fixed significance level of, for example, 0.05, the Type I error rate is kept constant. In contrast, for increasing sample sizes, the power can get very high (see Table 3), implying a decreasing probability of Type II errors (i. e. probability of not rejecting a false null hypothesis). Hence, the Type I error gets gradually prioritized over the Type II error, as the sample size increases.

The problem with a fixed significance level has long been recognized as a weakness (Schwab et al., 2011; Starbuck, 2016), and as large sample sizes become more common in research the issue gets worse. When using a fixed significance level, the Type I error probability can be orders of magnitude greater than the Type II error probability in large samples (Wulff & Taylor, 2024). Instead of using a fixed significance level when testing for MI, an alternative is to follow Mudge et al. (2012), and instead use a significance level that minimize the average probability of Type I and Type II errors, assuming the prior probabilities of the null and alternative hypothesis are equal.

Table 4 gives the significance levels which minimize the average

Table 4
Significance levels balancing probability of Type I and Type II errors.

Scenario	Difference in loadings between groups	Sample size in each group			
		100	*250*	*500*	*1000*
6 (Metric invariance)	0.05	0.460	0.314	0.247	0.223
	0.10	0.374	0.287	0.227	0.117
	0.15	0.233	0.183	0.120	0.030
	0.20	0.255	0.120	0.040	0.007
7 (Metric invariance, outer error correlation)	0.05	0.727	0.297	0.230	0.290
	0.10	0.424	0.263	0.207	0.117
	0.15	0.203	0.163	0.094	0.027
	0.20	0.230	0.107	0.040	0.007
	Difference in intercepts between groups				
8 (Scalar invariance)	0.10	0.277	0.227	0.123	0.060
	0.20	0.190	0.074	0.013	0.001
	0.30	0.080	0.013	0.001	0.001
	0.40	0.030	0.001	0.001	0.001
9 (Scalar invariance, outer error correlation)	0.10	0.243	0.137	0.053	0.010
	0.20	0.080	0.010	0.001	0.001
	0.30	0.016	0.001	0.001	0.001
	0.40	0.003	0.001	0.001	0.001

Note: The values in the columns under "Sample size in each group" show the significance level, α_{sig}, which minimizes $\frac{\alpha_{sig} + \beta_{err}}{2}$, where β_{err} is the probability of a Type II error for a given critical effect size (i.e. difference in loadings or intercepts). The metric (scalar) invariance test is based on the null hypothesis that there is no difference between the groups loadings (intercepts). $\beta_{err} = 1 - \frac{1}{R}\sum_{r=1}^{R} I[p_r \leq \alpha_{sig}]$ where R is the total number of Monte Carlo runs (6000), p_r is the p-value in r'th simulation, and $I[\cdot]$ is the indicator function. Significance levels of 0.001 indicate that, using the simulated data, the significance level balancing Type I and Type II error probabilities is not found, as these are too small (i.e. smaller than or equal to 0.001) probabilities of Type I and Type II errors for the power simulations in Table 3. The values in the column containing “difference in loadings between groups” and “difference in intercepts between groups” are the critical effect sizes and indicate the minimum effect the researcher believes has substantive importance. From Table 4, we see that the suggested significance level decreases as the sample size increases, and in general decrease as the critical effect sizes increases. Hence, researchers using larger sample sizes will often be more likely not to reject the null hypothesis of measurement invariance compared to using a fixed significance level of 0.05. A researcher following this approach, can use Table 4 as a guidance on the chosen significance level when evaluating MI with the proposed method, given their empirical setting resembles the simulation study. As the present simulation study is limited, larger simulation studies could support empirical researcher by covering a

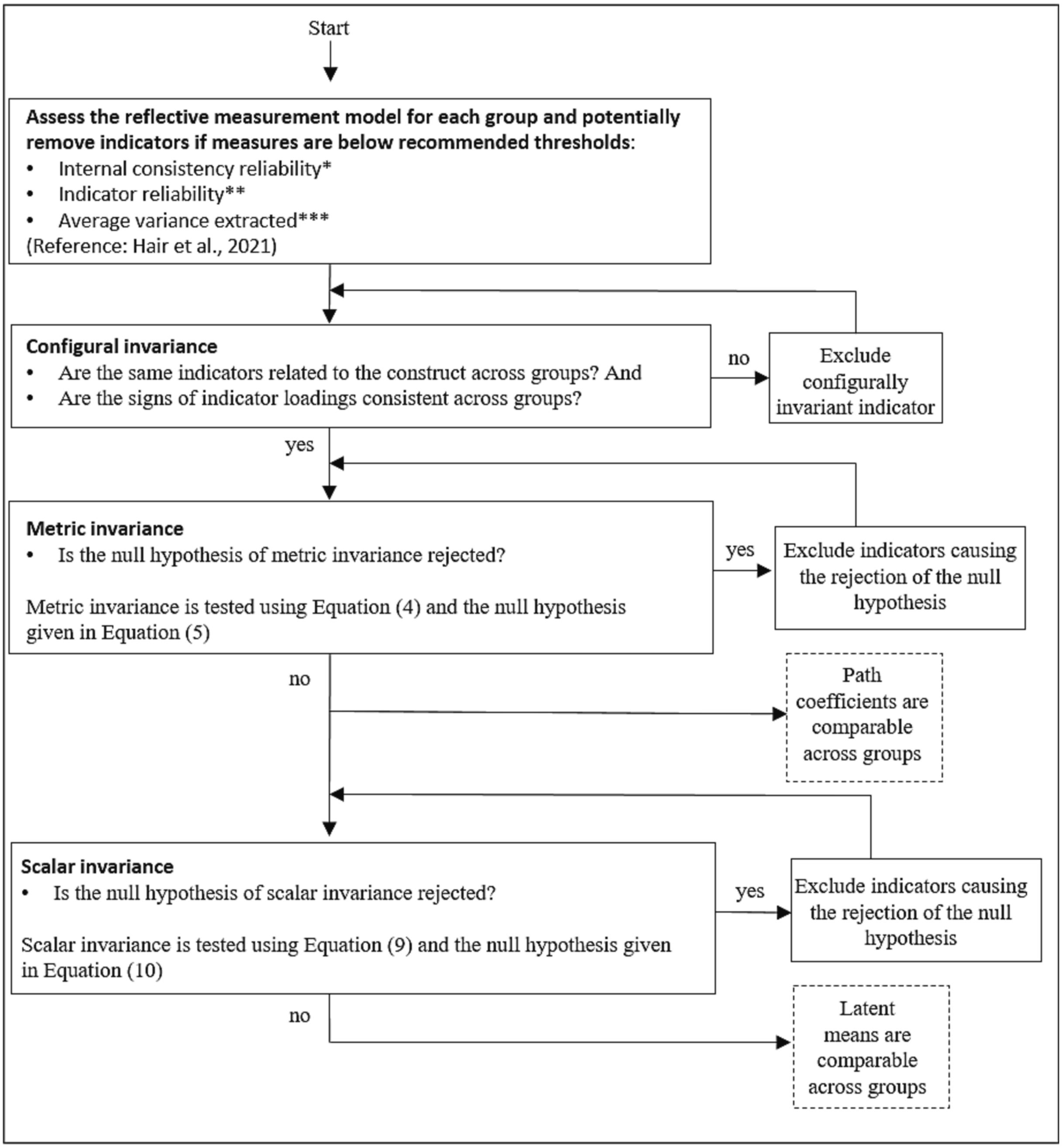

Fig. 3. Proposed procedure for assessing measurement invariance using PLS-SEM.
Note: When excluding indicators due to a failure to reject the null hypothesis of metric (scalar) invariance, consider the magnitude of loading (intercept) differences as guidance. Also, refer to the individual tests on loading (intercept) differences as outlined in Sections 4.1 (4.2).
The recommendations for assessing the reflective measurement model in PLS-SEM are described in detail in Hair et al. (2021), but a summary of the relevant measures are given below.
* Internal consistency reliability can be assessed using Cronbach's α, composite reliability, and the exact reliability coefficient. The recommended range of these measures are between 0.6 and 0.9 but depends on the research context.
** Loadings above 0.7 are retained, loadings between 0.4 to 0.7 is accepted depending on internal consistency reliability, and for loadings below 0.4 it is recommended to remove the indicator.
*** Average variance extracted is recommended to be above 0.5.

broader class of models and presenting their corresponding alternative significance levels.

6. Recommended procedure

Fig. 3 depicts a flowchart containing the proposed procedure for MI testing.[7] The general procedure of assessing levels of invariance sequentially is followed but note that this convention might lead to inflated Type I errors due to multiple comparisons (Nye & Drasgow, 2011).

The first step in the procedure is to evaluate the reflective measurement model in each group according to internal consistency reliability, indicator reliability, and average variance extracted, following the guidelines given in Hair et al. (2021). This step involves the removal of indicators with close to zero loadings in each group. Next, it is investigated whether the latent variables are configural invariant across groups. There is configural invariance if the same indicators are related to the same constructs across groups, and the signs of indicator loadings are consistent across groups. The subsequent tests are only conducted on latent variables that signal configural invariance.

When testing for metric invariance, the bootstrapped significance test, as described in relation to Equation (6), is used. If the test of metric invariance rejects the null hypothesis of equal indicator loadings across groups, one can consult the magnitude of the coefficients estimated from Equation (4). Large coefficients will guide us in pinpointing the indicators responsible for the MI violation, and these indicators may be eligible for removal. The overall test of metric invariance can be supplemented with tests for individual indicators, as outlined in Section 4.1. In the case of metric invariance, one can – with reasonable certainty – say that the construct is comparable across groups and that indicators are equally reliable.

If a researcher wishes to compare latent variable means across groups, the next step is to test for scalar invariance. If scalar invariance is rejected for like constructs, identifying large coefficients from estimating Equation (9) can point to the problematic indicators. Such inspections can be supplemented with tests for scalar invariance for individual indicators, as outlined in Section 4.2. If researchers can establish scalar invariance, then they have support for making meaningful comparisons of the latent means across groups.

While omitting specific indicators that cause a lack of measurement invariance can facilitate comparisons of structural coefficients and latent means between groups, such omissions might pose challenges for future research when comparing constructs across different studies. Therefore, it is recommended that researchers present model results both for each group individually (before potential deletion of indicators) and for the results of MI testing. If the goal is solely to test if constructs are measured in the same way across groups, without a subsequent comparison of structural coefficients or latent means, then there is no need to remove indicators.

7. Example: Two independent groups with two repeated measures

To facilitate the use of the proposed procedure and show how this method allows researchers to analyze experimental survey designs using PLS-SEM, this section provides an example which illustrates the steps of MI testing. This example investigates the effect of availability of information about data breaches on individuals' perceived online shopping risk (see Alrawad et al. (2023) for a cross-sectional analysis of customer perception of online shopping risk). The availability of information is essential for individual risk perception (Kahneman, 2011; Van Schaik et al., 2017), and the effect of perceived risk is in turn a relevant factor for understanding consumer decision-making (Meents & Verhagen, 2018). Consequently, the author of this study hypothesizes that individuals who obtain information about data breaches will change their risk perception of doing online shopping. Testing this hypothesis can help us to understand how, for example, news about data breaches can affect online consumer behavior. A 2 x 2 mixed design is used to estimate the effect of availability of information about data breaches on individuals' perceived online shopping risk.

The data were collected using an online survey sampled in the US and yielded 538 valid responses. The data collection was facilitated using Amazon Mechanical Turk, while taking the point of departure in the guidelines by Lowry et al. (2016).[8] First, the respondents were asked to answer questions measuring their perceived online shopping risk (see Appendix A.2) using a scale from previous research (Chakraborty et al., 2016). Second, based on random splitting, the respondents were divided into a control and a treatment group. The members of the treatment group were given information about data breaches in the US and were asked to indicate the degree to which this information was familiar to them. The control group was given information about heart disease and cardiovascular diseases and was also asked to indicate their level of knowledge about the information. The wording in the information provided to the control and treatment group was similar, and both types of information can be viewed as negative. Lastly, the respondents were asked to fill out the perceived online shopping risk scale again (i.e. to answer the same questions they answered before they were given information about either data breaches or heart and cardiovascular disease). Hence, the control and treatment group are created by priming them (Tversky & Kahneman, 1973) with different information.

Introducing a treatment in an experimental study might change the meaning of the construct (Chen, 2008; Cheung & Rensvold, 2002) and to address this concern it is investigated whether there is measurement invariance. The sampling scheme described above allowed for a PLS-SEM estimation in both the treatment group and the control group (see Fig. 4) and subsequently uses the composite scores to test the different levels of measurement invariance.

The perceived online shopping risk is measured four times (twice for the control group and twice for the treatment group). In the following, perceived online shopping risk is treated as one construct and MI is tested across the four measurements: pre-intervention for the control group; post-intervention for the control group; pre-intervention for the treatment group; and post-intervention for the treatment group. Given that respondents are asked the same questions twice, there is potential for correlated outer errors in the measurement model. Consequently, panel bootstrapping is employed for inference.

7.1. Assessing the reflective measurement models and configural invariance

To estimate the two models given in Fig. 4, the following PLS-SEM settings are used: equal weights for initialization of the algorithm (Hair et al., 2017a), correlation weights for the outer model estimation (Chin, 1998; Wold, 1982), path weighting scheme for the inner model estimations (Lohmöller, 1989), and a stop criterion of 10^{-7} (Tenenhaus et al., 2005). The SmartPLS 4 software (Ringle et al., 2022) is utilized to assess the reflective measurement model in each group. The levels of internal consistency reliability, indicator reliability, and average variance extracted are found to be satisfactory (see Appendix A.3). Furthermore, as the same indicators are related to the POSR construct

[7] The R code (R Core Team, 2022) to evaluate metric and scalar invariance according to the procedure in this paper, as well as an example, is available at https://github.com/ECONshare/Measurement-invariance-PLS-SEM.

[8] The sample is restricted to respondents who had approval rates of more than 90% on previously completed tasks on Amazon Mechanical Turk; only US respondents are sampled and the respondent's location is subsequently verified through their IP address; an attention question is included and the respondents are offered a pay equivalent to the federal minimum wage.

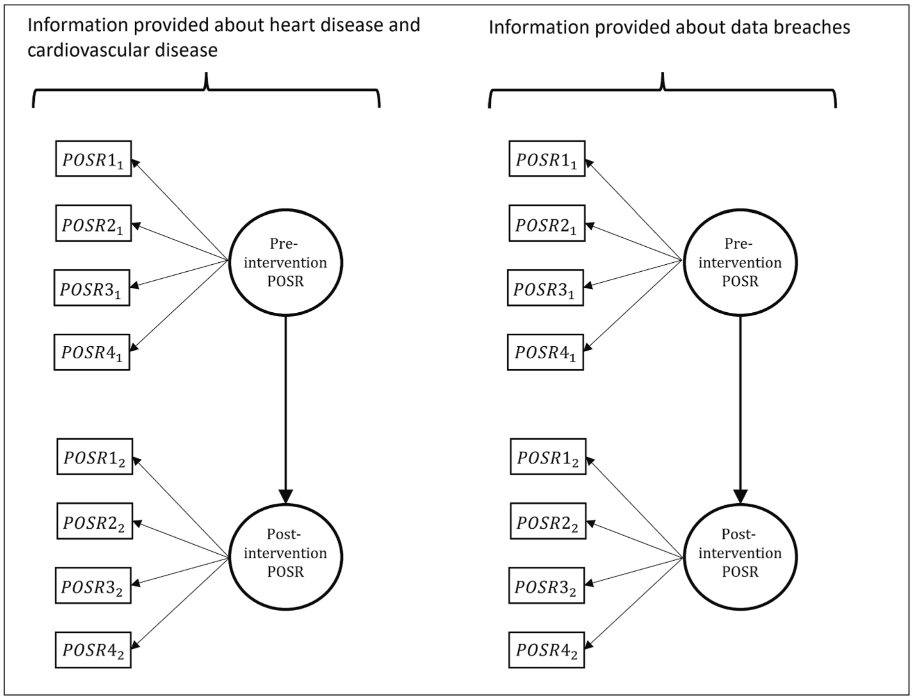

Fig. 4. Structural equation model for two groups in two time periods.
Note: The two identical SEMs represent two distinct groups: the control group (left) and the treatment group (right). POSR1, POSR2, POSR3, and POSR4, denote the four questions/indicators used to measure perceived online shopping risk (POSR). A subscript of "1" reflects that the question is answered prior to providing information to the respondents, while "2" denotes post-information responses.

across groups, and given the loadings all have the same sign, there is evidence of configural invariance.

7.2. Metric invariance

In testing for metric invariance, the pre-intervention measure of perceived online shopping risk in the control group is used as the reference group. Table 5 reports the coefficients estimated from Equation (4). Under the joint scaling of POSR over the groups (see Section 4.1), we can interpret the coefficient of, for example, POSR1 in the pre-intervention treatment group as the difference between this group's loading on POSR1 and the loading in the reference group (pre-intervention control group) on POSR1.

Due to repeated measures, panel bootstrapping (5000 resamples) is utilized. The null hypothesis is that all differences in loadings are zero and the corresponding test yielded a p-value of 0.401. Hence, we cannot reject the null hypothesis of metric invariance using a 0.05 significance level.[9] These findings suggest that we can proceed to testing for scalar invariance while restricting the loadings to equality between groups.

7.3. Scalar invariance

In testing for scalar invariance, the pre-intervention control group is kept as the reference group, and POSR3 is used as the marker indicator. Table 6 reports the estimated coefficients from the scalar invariance testing. The panel bootstrapped p-value is 0.026, implying that the null hypothesis of scalar invariance is rejected using a 0.05 significance level.[10] From Table 6 it is evident that POSR4 has a relatively high value in the pre-intervention treatment group[11] and the indicator is removed from the model. The resulting model without POSR4 gives a p-value of 0.061 and one cannot reject the null hypothesis of scalar invariance. Hence, we can proceed to what the author calls a "latent ANCOVA" model where the latent means are compared across groups. The latent ANCOVA is conducted without POSR4 as it was found to be variant across groups.

7.4. Latent ANCOVA model

The preceding MI tests provide confidence that the same construct is measured in all four measurements, after removing POSR4 from the model. To investigate whether the availability of information about data breaches affects individuals' perceived online shopping risk, an ANCOVA model is estimated using derived latent variable scores. Because the MI tests supported comparability of perceived online shopping risk across the treatment and control groups, the latent variable scores are estimated by conducting a PLS-SEM estimation on the pooled data.

Using the estimated latent variable scores, the following ANCOVA

[9] Referring to Table 4 to consider an alternative significance level, we can observe the following: given a critical effect size of 0.1, a sample size of approximately 250 in each group, and the potential for correlated outer errors, the appropriate significance level to evaluate the metric invariance test is 0.263. A p-value of 0.401 aligns with the conclusion drawn using the conventional 0.05 significance level; namely, one cannot reject the null hypothesis of metric invariance.

[10] Referring to Table 4 to consider an alternative significance level and choosing a critical effect size of 0.2 with a sample size of approximately 250 in each group, and the potential for correlated outer errors, the appropriate significance level to evaluate for scalar invariance test is 0.010. With a p-value of 0.026, one would not reject the null hypothesis of scalar invariance. However, with a critical effect size of 0.1, one would evaluate on a significance level of 0.137 and reject the null hypothesis of scalar invariance. Given that the conclusion is sensitive to relatively small changes in the critical effect size, the conclusion using the conventional 0.05 level is maintained.

[11] Testing only POSR4 for scalar invariance yields a *p*-value below 0.001. Hence, the null hypothesis that POSR4 is invariant is rejected, and supports the removal of the indicator.

Table 5
Test for metric invariance.

	Pre-intervention							
	Control group (and reference group)				Treatment group			
Indicator	POSR1	POSR2	POSR3	POSR4	POSR1	POSR2	POSR3	POSR4
Δ loadings	–	–	–	–	0.010	−0.003	0.043	−0.270
	Post-intervention							
	Control group				Treatment group			
Indicator	POSR1	POSR2	POSR3	POSR4	POSR1	POSR2	POSR3	POSR4
Δ loadings	0.016	0.071	0.007	−0.041	−0.088	−0.066	0.055	−0.165

Note: The p-value (0.401) is under the null hypothesis of metric invariance between the following groups; pre-intervention control group, post-intervention control group,pre-intervention treatment group, and post-intervention treatment group. With the pre-intervention control group as the reference group, its values are omitted, while the other values represent loading differences (Δloadings) relative to it. The reported Δloadings correspond to the estimated values of the vector, π_k^D as given in section 4.1. the Significance testing is based on 5000 panel bootstrap resamples.

Table 6
Test for scalar invariance.

	Pre-intervention							
	Control group (and reference group)				Treatment group			
Indicator	POSR1	POSR2	POSR3	POSR4	POSR1	POSR2	POSR3	POSR4
Δ intercepts	–	–	–	–	0.184	−0.011	Marker	0.206
	Post-intervention							
	Control group				Treatment group			
Indicator	POSR1	POSR2	POSR3	POSR4	POSR1	POSR2	POSR3	POSR4
Δ intercepts	−0.129	0.171	Marker	−0.045	−0.058	0.151	Marker	−0.043

Note: The p-value (0.027) is under the null hypothesis of full scalar invariance between the following groups; pre-intervention control group, post-intervention control group, pre-intervention treatment group, and post-intervention treatment group. With the pre-intervention control group as the reference group, its values are omitted, while the other values represent intercept differences (Δintercept) relative to it. The reported Δintercept correspond to the estimated values of the vector α_k^D, as given in Section 4.2. POSR3 is the marker variable, and its intercepts are assumed equal across groups. Significance testing is based on 5000 panel bootstrap resamples.

model is specified,

$$POSR_{post} = \mu + \kappa d + \beta_1 \left(POSR_{pre} - \overline{POSR}_{pre}\right) + \beta_2 INT + \zeta, \tag{12}$$

where μ is an overall intercept, d is a dummy variable equal to one if the observation belongs to the treatment group and zero otherwise. $POSR_{pre}$ and $POSR_{post}$ are the pre-intervention and post-intervention measures of perceived online risk. κ measures the effect of being in the treatment group when $POSR_{pre}$ is at its average ($\overline{POSR}_{pre}$), β_1 measures the relation between pre- and post-intervention common to both the treatment and control group, $INT = \left(POSR_{pre} - \overline{POSR}_{pre}\right)d$ is an interaction term, β_2 measures the difference between the treatment and control groups with respect to the relation between pre- and post-intervention, and ζ is the error term.

The results are given in Table 7, with the p-values based on 5000 bootstrap resamples. The significant effect of INT suggests that the relation between the pre-intervention measure of perceived online shopping risk and the post-intervention measure is lower for the treatment group. The coefficient on the treatment group dummy is 0.315 and highly significant. This suggests that risk perception increases as information about data breaches is provided. Both the control and treatment groups received information that can be perceived as negative, so the increased risk perception should be due to the substantive content of the information and not related to its emotional effect.

Table 7
Latent ANCOVA.

Dependent variable: Post-intervention POSR		
Variable	Coefficient	P-value
Intercept	4.300	0.00
Treatment dummy (d)	0.315	0.00
$POSR_{pre} - \overline{POSR}_{pre}$	0.861	0.00
INT	−0.148	0.019

Note: The table presents results from estimating Equation (12), using latent variable scores derived from PLS-SEM with pooled observations in a model equivalent to the models shown in Fig. 4, excluding POSR4. P-values are determined from 5000 bootstrap resamples.

8. Limitations and future research

This study, (1) synthesized the current MI testing literature into a PLS-SEM framework, (2) introduced a test for metric and scalar invariance that can be performed in case of unequal group sizes, an arbitrary number of groups, and outer error correlation, (3) found, through simulations, that the tests for metric and scalar invariance have adequate Type I error rates and power levels, (4) proposed a strategy to handle the problem of measurement invariance rejections in studies with large samples, (5) provided guidelines on how to sequentially perform the different levels of MI tests, and (6) by means of an empirical example, illustrated how the proposed procedure allows researchers to perform experimental survey designs within a PLS-SEM framework. While the study makes several contributions, beyond the traditional MICOM method, it has limitations and opens the avenue for future research. It is worth noting that there is a large overlap between the limitations and suggested future research outlined below and what are currently unresolved issues in CB-SEM. As this study aligns MI testing in PLS-SEM and CB-SEM to a greater extent than previous literature, researchers might benefit from new developments in tests for MI across the two types of estimation methods.

Like all significance testing procedures, the procedure proposed in this study is sensitive to the sample size and will tend to reject the null hypothesis of invariance more often as the sample size increases. This study proposes to accommodate the problem by employing the method of Mudge et al. (2012), which balances Type I and Type II errors. However, to determine the critical effect sizes (the minimum loading or intercept differences that are considered important to detect) using this method, it would be beneficial to investigate how large the difference in loadings or intercepts between groups should be before we ought to be concerned about the conclusions of the MI tests.

The effect of choosing non-invariant marker indicators needs further investigation in CB-SEM (Vandenberg & Morelli, 2016), and it would also be beneficial to study this in terms of the procedure outlined in this paper.

Depending on the level of MI tested, constraints are imposed on the parameters between groups in the testing equations, but these constraints are not imposed in the PLS-SEM estimation of the latent variable scores. In system estimation (Cameron & Trivedi, 2005), for instance, there can be efficiency gains by estimating a model with imposed constraints. This would likely also be true for PLS-SEM, and more efficient parameter estimates would in turn lead to higher power levels (Cameron & Trivedi, 2005). Hence, future research could investigate ways to incorporate parameter constraints into the PLS-SEM algorithm, which in turn would also make it possible to better align PLS-SEM estimation with CB-SEM frameworks, such as mean and covariance structure analysis (Ployhart & Oswald, 2004).

The procedure proposed in this paper might be used as a complement to the MI testing otherwise conducted in a CB-SEM context. In particular, future research could compare this study methodology to CB-SEM with respect to MI tests with many groups,[12] non-normality of the outer errors, low sample size, etc.

While the simulation study focuses on a two-construct model with a fixed number of indicators, there is ample opportunity for future explorations. To gain a more comprehensive understanding of the metric and scalar invariance tests' performance, future studies could expand the simulation designs. Such investigations could consider varying the complexity of the inner model, models with interaction terms and/or higher-order constructs, changing the number of indicators, testing across many groups, using different types of indicator distributions, and model misspecifications.

The framework outlined in this paper only needs the derived latent variable scores and the indicators for MI testing. Future research could therefore investigate the performance when using other SEM techniques that produce latent variable scores, such as generalized structured component analysis (Jung et al., 2012) and regularized generalized canonical correlation analysis (Tenenhaus et al., 2015; Tenenhaus et al., 2017).

9. Conclusion

When a business researcher wishes to compare structural equation models over time for the same individuals or between groups of individuals, the conclusions are only meaningful if the invariance of the measured constructs is established. A lack of MI suggests that the construct carries different meanings across groups or occasions, which can result in misleading comparisons. This study proposes a procedure that enables the researcher to test the credibility of comparing path coefficients and latent means across groups in PLS-SEM. In a longitudinal setting, the method is capable of metric and scalar invariance testing with an arbitrary number of equal-sized groups and possibly correlated outer measurement errors. If correlated outer errors are not expected, the proposed tests in addition allow unequal group sizes. Through simulations it is found that the proposed tests of metric and scalar invariance testing show satisfactory Type I error rates and power levels. As a solution to the common problem of rejecting MI in large-sample studies, this paper proposes the use of alternative significance levels in accordance with Mudge et al. (2012).

The proposed framework enables researchers to perform new types of analysis in PLS-SEM, including experimental designs and latent mean comparisons, capabilities not supported by the widely used MICOM approach.

CRediT authorship contribution statement

Benjamin Dybro Liengaard: Conceptualization, Data curation, Visualization, Investigation, Validation, Formal analysis, Methodology, Project administration.

Declaration of competing interest

The author declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A

Appendix A.1

Without a loss of generality, Group 1 will be the reference group and the first indicator, $k = 1$, is the marker indicator. For $g = 1, \ldots, G$ and $k^* = 2, \ldots, K$, it will be shown how the coefficient vectors from the equation used to test for scalar invariance (Equation (9)) are functions of the intercepts in the reflective measurement model (Equation (2)). More specifically, by considering the coefficient vectors $\alpha^A_{k=k^*} = \left[\alpha^r_{g=1,k=k^*}, \alpha^d_{g=2,k=k^*}, \ldots, \alpha^d_{g=G,k=k^*}\right]'$ given in Equation (9), it is first shown that the coefficients $\alpha^r_{g=1,k=k^*}$, are functions of the intercepts $\alpha_{g=1,k=k^*}$, and secondly shown that the remaining entries $\alpha^D_{k=k^*} = \left[\alpha^d_{g=2,k=k^*}, \ldots, \alpha^d_{g=G,k=k^*}\right]'$ are the difference in the intercept between Group 1 and Group $g^* = 2, \ldots, G$.

The use of Group 1 as reference group is reflected in the dummy structure of I^M (see Equation (7)). This dummy structure in turn implies that $\alpha^r_{g=1,k=k^*}$ is equal to the expected value of $\lambda_{g=1,k=k^*}$ when Equation (9) is estimated with OLS. Hence,

[12] For example, in the framework of mean and covariance structure analysis (Ployhart & Oswald, 2004) the estimation becomes unwieldy if there are about five groups or more.

$$
\begin{aligned}
\alpha^{r}_{g=1,k=k^*} &= E\left[\lambda_{g=1,k=k^*}\right] \\
0 &= E\left[x_{g=1,k=k^*} - \frac{\pi^{res}_{k=k^*}}{\pi^{res}_{k=1}} x_{g=1,k=k^*}\right] \\
0 &= \alpha_{g=1,k=k^*} + \pi^{res}_{k=k^*} E\left[Y_{g=1}\right] - \frac{\pi^{res}_{k=k^*}}{\pi^{res}_{k=1}} \alpha_{g=1,k=k^*} - \pi^{res}_{k=k^*} E\left[Y_{g=1}\right] \\
0 &= \alpha_{g=1,k=k^*}\left(1 - \frac{\pi^{res}_{k=k^*}}{\pi^{res}_{k=1}}\right),
\end{aligned}
$$

for $k^* = 2, \dots, K$.

Due to the dummy structure of I^M, each entry in the vectors $\alpha^{D}_{k=k^*} = \left[\alpha^{d}_{g=2,k=k^*}, \dots, \alpha^{d}_{g=G,k=k^*}\right]'$, measures the difference between the expected value of $\lambda_{g=1,k=k^*}$ and the expected value of each of the vectors $\lambda_{g=2,k=k^*}, \lambda_{g=2,k=k^*}, \dots, \lambda_{g=G,k=k^*}$. For each $g^* = 2, \dots, G$ and $k^* = 2, \dots, K$ we therefore have

$$
\begin{aligned}
\alpha^{d}_{g=g^*,k=k^*} &= E\left[\lambda_{g=g^*,k=k^*}\right] - E\left[\lambda_{g=1,k=k^*}\right] \\
0 &= E\left[x_{g=g^*,k=k^*} - \frac{\pi^{res}_{k=k^*}}{\pi^{res}_{k=1}} x_{g=g^*,k=1}\right] - E\left[x_{g=1,k=k^*} - \frac{\pi^{res}_{k=k^*}}{\pi^{res}_{k=1}} x_{g=1,k=1}\right] \\
0 &= \alpha_{g=g^*,k=k^*} + \pi^{res}_{k=k^*} E\left[Y_{g=g^*}\right] - \frac{\pi^{res}_{k=k^*}}{\pi^{res}_{k=1}} \alpha_{g=g^*,k=1} - \pi^{res}_{k=k^*} E\left[Y_{g=g^*}\right] \\
0 &- \alpha_{g=1,k=k^*} - \pi^{res}_{k=k^*} E\left[Y_{g=1}\right] + \frac{\pi^{res}_{k=k^*}}{\pi^{res}_{k=1}} \alpha_{g=1,k=1} + \pi^{res}_{k=k^*} E\left[Y_{g=1}\right] \\
0 &= \alpha_{g=g^*,k=k^*} - \alpha_{g=1,k=k^*}
\end{aligned}
$$

where the last equality uses the marker variable assumptions that $\alpha_{g=g^*,k=1} = \alpha_{g=1,k=1}$.

Appendix A.2

Perceived online shopping risk	
The control group answered the questions POSR1 to POSR4, then answered the control question, and finally answered questions POSR1 to POSR4 again. The treatment group answered the questions in the same order, but instead of the control question, they answered the treatment question.	
Scale for POSR1 to POSR4 (1 = Fully Disagree to 7 = Fully Agree). Scale for control and treatment questions (“Yes”, “I had an idea”, or “No”)	
POSR1	Online shopping websites are vulnerable to hackers who may steal customers’ information.
POSR2	Online shopping customers often face damaging and harmful behavior from hackers.
POSR3	Customers’ information stored at online shopping websites is not safe.
POSR4	Customers are vulnerable in online shopping websites that had incidents of hacking.
Control	Did you know that 27.6 million American adults have been diagnosed with heart disease and that the direct and indirect cost of cardiovascular disease in the US in 2018 was 329.7 billion?
Treatment	Did you know that due to data breaches in the US in 2018, 446.52 million American records were exposed, and that the reported monetary damage was 2.71 billion?

Appendix A.3

Measurement model					
		Control group		Treatment group	
		Pre-intervention	Post-intervention	Pre-intervention	Post-intervention
OuterLoadings	POSR 1	0.791	0.836	0.809	0.861
	POSR 2	0.712	0.780	0.680	0.769
	POSR 3	0.789	0.767	0.800	0.840
	POSR 4	0.836	0.826	0.740	0.790
Cronbach’s α		0.788	0.816	0.754	0.832
ρ_A		0.794	0.818	0.762	0.838
Composite reliability		0.863	0.879	0.844	0.880
Average Variance Extracted (AVE)		0.613	0.644	0.576	0.665

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